

Sharone Abhilash

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Education

Specialist in Astrophysics, Minor in Computer Science & Mathematics

June 2027

University of Toronto

cGPA: 3.7/4.0

Relevant Coursework: Observational Astronomy (AST325) · Astrophysics I & II (AST221/222) · Stellar Astrophysics (AST320) · Mathematical & Computational Physics (PHY325) · Quantum Mechanics (JCP321) · Statistical Mechanics (JCP322) · Partial Differential Equations (MAT311) · Numerical Methods (CSC338) · Optics (PHY347) · Intro to Computer Programming (CSC108/148) · Supervised Reading (PHY473, AST430) · Senior Thesis (CPS489)

Skills

Python (NumPy, Pandas, SciPy, Scikit-learn, HDBSCAN, Astropy, Galpy, Emcee, joblib) · MESA Stellar Evolution Code · Science Communication · Critical Thinking · Adaptability · Persistence · Time Management · Teamwork & Collaboration

Research Projects & Positions

Investigate Planet Properties Across The Exo-Neptune Landscape

Ongoing Research Project/Position at UofT

- Analyzed host-star metallicities, eccentricities, and density distributions across Neptune subpopulations.
- Applied MESA stellar evolution code and custom modifications to simulate planet evolution and constrain core masses across the Exo-Neptune landscape.
- Investigated trends with metallicity and irradiation level.

Star Formation in ρ Ophiuchi

Dunlap Institute for Astronomy & Astrophysics; Ongoing, Planned Publication

- Developed a multigenerational picture of star formation by integrating SPYGLASS pre-main-sequence stars, protostars, prestellar dense cores, Herbig-Haro objects, and gas maps.
- Studied spatial patterns, age gradients, and dynamics as they relate to feedback-driven star formation and regional expansion.

Herbig-Haro Objects in Ophiuchus

Dunlap Institute for Astronomy & Astrophysics; Ongoing, Planned Publication

- Catalog and characterize Herbig-Haro objects in ρ Ophiuchi in the context of their protostellar progenitors.
- Investigated the role of HH objects in quenching star formation by analyzing spatial correlations between HH object positions and the temperatures and locations of starless dense cores.

Publications

Nth Author

- [1] Rector, T.A., Kerr, R.M.P., Prato, L., Shuping, R.Y., Bender, C., Esplin, T.L., & **Abhilash, S.E.** 2026, Mapping Active Star-Formation in Serpens and the Aquila Rift, *AJ*, doi:10.48550/arXiv.2606.05668

Work & Volunteer Experience

Astronomy Member at Large / *Erindale Chemical and Physical Sciences Society*

June 2026 – Present

- Advertise ECPSS events within astronomy classrooms to increase interdepartment engagement.
- Represent Astronomy at the Chemical and Physical Sciences Department in the ECPSS.

External Affairs Manager / *UTM Astronomy Student Association*

December 2024 - Present

- Lead planning and coordination of academic, outreach, and networking events for undergraduate astronomy students.
- Support student engagement by connecting members with research, mentorship, and professional opportunities.
- Monitor current astronomy events to keep students informed and plan programming accordingly.

Astronomy Researcher / *Space Point Publications*

February 2025 - April 2026

- Conducted in-depth analyses of scientific literature to support astronomy and physics content for magazine issues, blog posts, and online courses.
- Provided research-backed insights to writers ensuring accuracy and engagement.

Volunteer Barista / *The Carrot Community Coffee House*

January 2021 - December 2022

- Maintained cleanliness, organized spaces, and ensured seamless operations during busy periods.
- Demonstrated teamwork and creativity while building connections with the local community and volunteers through art and coffee.

Awards & Accomplishments

- SURP 2025 Professor Mercedes T. Richards Award · *Honourable Mention*
- SURP 2025 Poster Session Award
- SURP 2025 Undergraduate Research Program, University of Toronto (\$10,881)
- University of Toronto Dean's List (2023/24, 2024/25, 2025/26)
- CKC Post Secondary Academic Award (2025, 2026)
- India Day Academic Achievement Award (2019–20)
- Academic Honours (2017–18, 2018–19, 2019–20, 2020–21)